

$$R_{xy} = \begin{pmatrix} c\phi & -s\phi & 0 \\ s\phi & c\phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \approx \mathbb{I} - i\phi T_z$$

$$T_z = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$R_{yz} \dots \quad T_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \quad T_y = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}$$

$$t_+ = t_x + it_y, \quad t_- = t_x - it_y$$

$$[t_z, t_+] = t_+, \quad [t_z, t_-] = -t_-$$

$$[t_+, t_-] = 2t_z$$

$$\text{* Tensors: } T(e_i \otimes e_j) = T e_i \otimes e_j + e_i \otimes T e_j$$

* Matrix Lie Groups:

$$GL(n, \mathbb{K}) = \{A \in M_n(\mathbb{K}) \mid \det(A) \neq 0\}$$

$$SL(n, \mathbb{K}) = \{A \in M_n(\mathbb{K}) \mid \det(A) = 1\}$$

$$O(n) = \{A \in M_n(\mathbb{R}) \mid A A^T = -I_n\}$$

$$O(p, q) = \{A \in M_n(\mathbb{R}) \mid A^T D_{p,q} A = D_{p,q}\}$$

$$D_{p,q} = \begin{pmatrix} \underbrace{1 \dots 1}_p & & \\ & \underbrace{1 \dots -1}_q & \\ & & \dots \end{pmatrix}$$

$$U(n) = \{A \in M_n(\mathbb{C}) \mid A A^T = I_n\}$$

$$U(p, q) = \{A \in M_n(\mathbb{C}) \mid A D_{p,q} A^T = D_{p,q}\}$$

$$SO(n) = O(n) \cap SL(n, \mathbb{R})$$

$$SU(n) = U(n) \cap SL(n, \mathbb{C})$$

Symplectic groups

$$w(x, y) \equiv \sum_{j=1}^n x_j y_{n+j} = x_{n+j} \neq i$$

A s.t.

$$w(Ax, Ay) = w(x, y)$$

$$\Rightarrow A \in Sp(n, \mathbb{R}) \subset GL(2n, \mathbb{R})$$

$$w(x, y) = \langle x, \Omega y \rangle \quad \Omega = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$$

$$-\Omega A^T \Omega = A^{-1}$$

or $M^T \Omega M = \Omega \quad | \quad \Omega^T = -\Omega$

" $\Delta M \Delta = n$ "

Euclidean group:

$$ISO(n) \equiv \left\{ \begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix} \in GL(n+1, \mathbb{R}) \mid A \in O(n) \right\}$$

$$v \equiv \begin{pmatrix} \vec{x} \\ 1 \end{pmatrix}$$

$$M.v = \begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \vec{x} \\ 1 \end{pmatrix} = \begin{pmatrix} A\vec{x} + b \\ 1 \end{pmatrix}$$

Heisenberg group:

$$H \equiv \left\{ \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} \in GL(3, \mathbb{R}) \right\}$$

$$[q, p] = i\hbar$$

$$U(a, b, c) \equiv e^{ic} e^{ib\hat{x}} e^{ia\hat{p}}$$

$$U_{a+b, c+c'} = U_{a+c', b+b', c+c'}$$

$$c \sim c + 2\pi n$$

• normal subgroup of H

$$Z \equiv \left\{ \begin{pmatrix} 1 & 0 & 2\pi n \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \mid n \in \mathbb{Z} \right\}$$

$$\begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & a & c+2\pi n \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 2\pi n \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$$

$H/\mathbb{Z}$: set of all related el. via $\mathbb{Z}$:

$$SO(2) \cong U(2) \cap SL(2) \cong U(1)$$

$$U(1) = \{ z \in \mathbb{C} \mid |z|^2 = 1 \}$$

$$SO(2) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} = A \mid \begin{matrix} ad - bc = 1 \\ AA^T = \mathbb{I} \end{matrix} \right\}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^2 + bc & ba + bd \\ ac + dc & bc + d^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$a^2 + bc = 1$$

$$b = c = 0$$

$$a = \pm 1, \quad d = \pm 1$$

$$SO(2) : \theta \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Definition: $e^{i\theta} = \cos(\theta) + i\sin(\theta)$

$$S^1 = \{ z \mid |z| = 1 \}$$

$$e^{i\theta_1} \cdot e^{i\theta_2} = e^{i(\theta_1 + \theta_2)}$$

$$R_\theta = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$$

$$SO(2) = \{ R_\theta \}$$

$$Z_\theta = (r=1) e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

$$\bullet (x, y) = x + iy$$

$$Z_\theta(x, y) = Z_\theta(x + iy) = (\cos(\theta) + i\sin(\theta))(x + iy)$$

$$= x\cos(\theta) - y\sin(\theta) + i(x\sin(\theta) + y\cos(\theta))$$

$$= (x\cos(\theta) - y\sin(\theta), x\sin(\theta) + y\cos(\theta))$$

• Exactly same as

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = R_\theta \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbb{Z}_\theta \xrightarrow{\sim} R_\theta$$

$$SO(2) \cong U(1) = S^1$$

* (Complex #'s) $\cong$ (Rotations in 2D)

See Th. 3:

$$R_\theta = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$$

$$R_\theta = \cos(\theta) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \sin(\theta) \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$\hat{I} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad | \quad \hat{L} \equiv \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$R_\theta = \cos(\theta) \hat{I} + \hat{L} \sin(\theta)$$

$$\bullet \hat{I}^2 = \hat{I}, \quad \hat{I}\hat{L} = \hat{L}\hat{I} = \hat{L}$$

$$\hat{L}^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -\hat{I}$$

$$\bullet \begin{pmatrix} a & -b \\ b & a \end{pmatrix} = a \hat{I} + \hat{L} b$$

$$\mathbb{C}: |a+bi|^2 = |a|^2 + |b|^2$$

$$(\text{Norm in } \mathbb{C}) \longleftrightarrow (\det \text{ in } SO(2))$$

$$\|\cdot\|: \mathbb{C} \rightarrow \mathbb{R}^+$$

$$\det: SO(2) \rightarrow \mathbb{R}$$

$$|z_1 z_2| = |z_1| |z_2|$$

$$\det(AB) = \det(A) \det(B)$$

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}^{-1} = \frac{1}{\det} \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

$$\text{SO}(2) \cong \text{U}(1)$$

Quaternions:

$$(a, b) \xrightarrow{\quad} a + ib \in \text{U}(1)$$

$$\begin{array}{c} \downarrow \\ \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \\ \eta \end{array}$$

$$\begin{array}{c} \nearrow \varphi^{-1} \\ \searrow \varphi \end{array}$$

$$\varphi(1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\varphi(i) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$\text{SO}(2)$$

Now work with 4-par

$$(a, b, c, d) \longrightarrow \dots$$

$$\downarrow$$

$$g = \begin{pmatrix} a+id & -b-ic \\ b-ic & a-id \end{pmatrix}$$

$$\text{SU}(2) \cong \text{S}^3$$

"

$$\left\{ \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix} \mid |a|^2 + |b|^2 = 1 \right\} \mid \det(M) = 1, M^T M = I \}$$

$$|a|^2 + |b|^2 = x_1^2 + x_2^2 + x_3^2 + x_4^2 = 1$$

$$\text{SO}(3): \{M \in \text{GL}(3, \mathbb{R}) \mid M^T M = I, \det(M) = 1\}$$

$$v \longrightarrow Mv$$

$$\langle v', v' \rangle = \langle v, v \rangle$$

$$\langle Mv, Mv \rangle = v^T v$$

$$(Mv)^T Mv = v^T M^T Mv = v^T v$$

- $V = V^\dagger \quad \text{and} \quad \text{tr}(V) = 0$

$$V = \begin{pmatrix} x & y+iz \\ y-iz & -x \end{pmatrix}$$

$$\det(V) = -x^2 + y^2 + z^2$$

* $V' = M^\dagger V M$:

- $V'^\dagger = V'$ ✓

- $\text{tr}(V') = 0$

- $\det(V') = \det(V)$

$$-M^\dagger V^* M = M^\dagger V M$$

$$SU(2)/\mathbb{Z}_2 \cong SO(3)$$

Symplectic Group:

$$\psi(i) = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \psi(j) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \psi(k) = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

$$\psi(1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\psi: \mathbb{H} \rightarrow GL(2, \mathbb{C})$$

$$\begin{aligned} \psi(\tilde{z}) &= \psi(\omega + ix + jy + kz) \\ &= \begin{pmatrix} \omega + ix & y + iz \\ -y + iz & \omega - ix \end{pmatrix} = q \end{aligned}$$

$$\psi(i)\psi(j) = \psi(k)$$

$$\psi_n: GL(n, \mathbb{H}) \rightarrow GL(2n, \mathbb{C})$$

q : closes under matrix mult.

$$|q|^2 = \det(q) = a^2 + b^2 + c^2 + d^2 \in \mathbb{R}^4$$

$$H = \langle 1, i, j, k \mid i^2 = j^2 = k^2 = -1, ij = k, ji = -k, jk = i, kj = -i, ki = j, ik = -j \rangle$$

$$q_1(q_2 q_3) = (q_1 q_2) q_3$$

$$qq^{-1} = 1 \quad | \quad q \neq 0$$

$$q1 = q$$

$$\begin{pmatrix} a+di & b-ci \\ b-ci & a-di \end{pmatrix} = a1 + bi + cj + dk$$

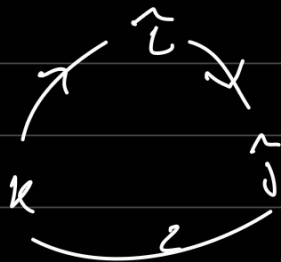
$$1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \hat{i} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$\hat{j} = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} \quad \hat{k} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

$$\hat{i} = -i\sigma_y = -i\sigma_2$$

$$\hat{j} = -i\sigma_x = -i\sigma_1$$

$$\hat{k} = i\sigma_z$$



$$ij = k$$

$$jk = i$$

$$ki = j$$

Norm on $\{q\}$

$$a, b, c, d \in \mathbb{R}$$

$$\det: \text{Matr}(H) \rightarrow \mathbb{R}$$

$$|q_1 q_2| = |q_1| |q_2|$$

$$\det(q_1 q_2) = \det(q_1) \det(q_2)$$

$$q^{-1} = \frac{1}{a^2 + b^2 + c^2 + d^2} (a1 - bi - cj - dk)$$

$\bar{q}$: Quaternionic conjugate:

$$\bar{1} = (a1 - bi - cj - dk)$$

$$q\bar{q} = a^2 + b^2 + c^2 + d^2 = |q|^2$$

$\bar{q}$: obtained from taking the complex conj. of each entry in the transposed matrix q^T

$$\begin{aligned}\bar{q} &= a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - c \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - d \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \\ &= \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} + \begin{pmatrix} 0 & b \\ -b & 0 \end{pmatrix} + \begin{pmatrix} 0 & ci \\ ci & 0 \end{pmatrix} + \begin{pmatrix} -di & 0 \\ 0 & di \end{pmatrix}\end{aligned}$$

$$\bar{q} = \begin{pmatrix} a - di & b + ci \\ -b + ci & a + di \end{pmatrix}$$

Note: $\bar{q} = q^T$

proof:

$$\begin{aligned}q &= \begin{pmatrix} a + id & -b - ic \\ b - ic & a - id \end{pmatrix} \\ q^* &= \begin{pmatrix} a - id & -b + ic \\ b + ic & a + id \end{pmatrix} \\ (q^*)^T &= \begin{pmatrix} a - id & b + ic \\ -b + ic & a + id \end{pmatrix} = q^T = \bar{q} \quad \checkmark\end{aligned}$$

$$\bar{q} \equiv q^T$$

$$V^\mu = \Theta^\alpha \sigma^\mu_{\alpha\dot{\alpha}} \bar{\Theta}^{\dot{\alpha}}$$

Thus: $(q_1 q_2)^T = \overline{q_1 q_2} = \bar{q}_2 \bar{q}_1$

3-sphere: S^3

• Unit - quaternions:

$$a^2 + b^2 + c^2 + d^2 = 1 \Rightarrow |q|^2 = 1$$

$$a^2 + b^2 + c^2 + d^2 = 1 \quad S^3 \subset \mathbb{P}^4$$

$S^3: (a, b, c, d)$ is a group under $\mathbb{H}$ mult

Purely imaginary $\mathbb{H}$:

$$p = bi + cj + dk \in \mathbb{R}^3$$

orthogonal complement of $\mathbb{R} \mathbf{1}$

"real quaternions"

• Not closed under mult.

$$u = u_1 i + u_2 j + u_3 k$$

$$v = v_1 i + v_2 j + v_3 k$$

$$\begin{aligned} uv = & -(u_1 v_1 + u_2 v_2 + u_3 v_3) \mathbf{1} \\ & + (u_2 v_3 - u_3 v_2) i - (u_1 v_3 - u_3 v_1) j \\ & + (u_1 v_2 - u_2 v_1) k \end{aligned}$$

• mult. $\mathbb{H}^c \subset \mathbb{R}^3 \subset \mathbb{H}$

$\hookrightarrow$ encodes $\mathbb{R}^3$ vector products

$$uv = -\langle \vec{u}, \vec{v} \rangle_{\mathbb{R}^3} + \vec{u} \times \vec{v}$$

* $\mathbb{H}^c$ closed iff $\langle \vec{u}, \vec{v} \rangle = 0$, i.e. u and v are orthogonal

$\mathbb{H}^c \rightarrow \mathbb{H}^{\mathbb{R}}$ if $\vec{u} \times \vec{v} = 0$ i.e. parallel in $\mathbb{R}^3$

$$* |u| = 1$$

$$u^2 = -u \circ u = -|u|^2 = -1$$

$\mathbb{H}$ rep of $\mathbb{R}^4$:

$$|t| = 1 \iff \det(t) = 1 = |t|^2$$

$$|t| = \sqrt{t \bar{t}}$$

$$t = \underbrace{\cos(\theta)}_{\in \mathbb{R}} \hat{I} + \underbrace{\sin(\theta)}_{\in \mathbb{R}} \hat{u}$$

u : unit vector in $\mathbb{H}^d$

$$|u| = 1$$

$$u^2 = -|u|^2 = -\hat{I} \quad | \quad |u|^2 = 1$$

$$\begin{aligned} t\bar{t} &= (\cos(\theta) \hat{I} + \sin(\theta) \hat{u})(\cos(\theta) \hat{I} - \sin(\theta) \hat{u}) \\ &= \cos^2(\theta) \hat{I} + \cancel{\sin(\theta) \hat{u}} - \cancel{\sin(\theta) \hat{u}} - \sin^2(\theta) \hat{u}^2 \\ &= (\cos^2(\theta) + \sin^2(\theta)) \hat{I} \end{aligned}$$

$$t\bar{t} = 1 = |t|^2$$

$$t^{-1} = \frac{\bar{t}}{|t|^2}$$

$$* \quad q \mapsto t^{-1} q t = \frac{1}{|t|^2} \bar{t} q t \quad (\text{Conjugation})$$

• isometry of $\mathbb{H}$:

$$t^{-1} \mathbb{R} \hat{I} t = \mathbb{R}$$

$$\mathbb{H}^{\mathbb{R}} \rightarrow \mathbb{H}^{\mathbb{R}}$$

$$\text{so } t^{-1} \mathbb{H}^c t \rightarrow \mathbb{H}^c$$

Thm: $t = \cos(\theta) \hat{I} + \hat{u} \sin(\theta) \quad \hat{u} \in \mathbb{H}^c$
 [Conj. by t rotates $\mathbb{H}^c$ through $|\hat{u}| = 1$
 an angle 2θ about the axis u]

proof:

$$\begin{aligned} t^{-1} u t &= (\cos(\theta) - \hat{u} \sin(\theta)) u (\cos(\theta) + \hat{u} \sin(\theta)) \\ &= (c_\theta u + s_\theta) (c_\theta + \hat{u} s_\theta) \\ &= c_\theta^2 u + s_\theta c_\theta + \hat{u} s_\theta^2 - s_\theta c_\theta \\ &= u \end{aligned}$$

$$t^{-1} u t = u$$

→ Isometry of $\mathbb{H}^c$: Thus

$$\mathbb{R} \hat{I} \rightarrow \mathbb{R} \hat{I}$$

if $t \in \mathbb{H}^d$, then

$$z = \vec{u} + \vec{u}_\perp$$

$$\bar{z} = \bar{u} + \bar{u}_\perp$$

$$z\bar{z} = (\vec{u} + \vec{u}_\perp)(\bar{u} + \bar{u}_\perp)$$

$$= 1 + \vec{u}_\perp \bar{u} + \vec{u} \bar{u}_\perp + \vec{u}_\perp \bar{u}_\perp + |\vec{u}_\perp|^2$$

$$\vec{u}_\perp \bar{u} = \vec{u}_\perp \times \vec{u} + \vec{u} \times \vec{u}_\perp$$

$$\stackrel{!}{=} 0$$

$$z\bar{z} = 1 + |\vec{u}_\perp|^2$$

$$\vec{u}_\perp : \vec{u}_\perp \cdot \vec{u} = 0$$

$$\omega \equiv \vec{u} \times \vec{u}_\perp$$

$$\text{Span}\{\vec{u}, \vec{u}_\perp, \omega\} \cong \mathbb{H}^d$$

$$\vec{u}_\perp \equiv V$$

$$\vec{u}\omega = \omega$$

$$\omega\vec{u} = \vec{u}$$

$$\vec{u}$$

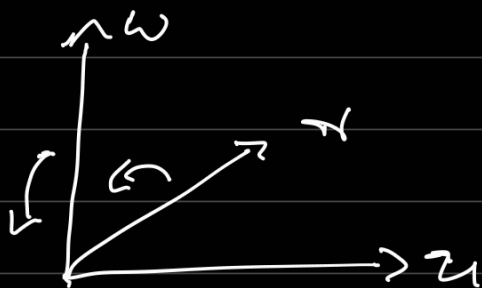
$$\omega\omega = \vec{u} \quad \vec{u}\omega = -\omega\vec{u}$$

$$V$$

$$\omega$$

$$\textcircled{a} t^{-1} \vec{u} t = \vec{u} \cos(2\theta) - \omega \sin(2\theta)$$

$$\textcircled{b} t^{-1} \omega t = \omega \cos(2\theta) + \vec{u} \sin(2\theta)$$



Ⓐ rotates V about $\vec{u}$ by 2θ

Ⓑ rotates ω about $\vec{u}$ by 2θ $\square$

$\rightarrow$ same rotation induced by

$$(-t^{-1}) \omega (-t) = t^{-1} \omega t$$

$\pm t$ are the only $\mathbb{H}^d$

$$t = \cos\left(\frac{\alpha}{2}\right) + 2i \sin\left(\frac{\alpha}{2}\right)$$

$$\bullet (u, \alpha) \text{ \& } (-u, -\alpha)$$

$\bullet t \text{ \& } -t$ are antipodal

* Rotations in $\mathbb{R}^3$ correspond to antipodal pairs of unit quaternions

Cor: Product of rotations is a rotation, and
 $\perp$ inverse of a rotation is a rot.

$$Sp(n) \equiv \{M \in GL(n, \mathbb{H}) \mid M^T M = I_n, \det(M) = 1\}$$

* Back to ex.